



Carbon dioxide removal and mitigation deterrence in EU climate policy: Towards a research approach^{★,☆☆,☆☆}

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ABSTRACT

Carbon dioxide removal (CDR) technologies are rapidly evolving from renderings in climate models towards an explicit and essential component of climate policy-making. This has raised concerns that the expectations of future large-scale availability of CDR technologies may deter mitigation, i.e., delay or substitute emission reduction efforts. Here, we develop a conceptual approach to systematically study such mitigation deterrence effects in climate policy-making processes. The approach integrates literature on sociotechnical imaginaries and the cultural political economy of mitigation deterrence with historical-materialist policy analysis. It follows within-case process tracing to uncover whether and how expectations of CDR technologies either weaken 'conventional' mitigation targets in policy formulation or encourage policy designs which treat 'conventional' mitigation and CDR as equivalent. Using critical instances of contestation in the policy process, this approach also allows us to identify nascent constellations of actors and interests as well as emerging fault lines related to CDR technologies. Empirically, the paper focusses on the EU Sustainable Carbon Cycles Strategy and the current negotiations over an EU Carbon Removal Certification Framework. We show that even though it is too early to pin-point clear-cut instances of mitigation deterrence related to CDR technologies in EU climate policy-making, critical opportunities to prevent such effects have already been missed in both climate target formulation and policy design.

1. Introduction

In the early days of international climate policy, it was still possible to think of climate change mitigation in terms of gradually reducing anthropogenic greenhouse gas (GHG) emissions. But as emissions have kept rising, this is no longer the case. Most climate science models suggest that limiting global warming to 1.5 °C or at least well below 2 °C requires not only rapid emission cuts but also large-scale removal of carbon dioxide from the atmosphere, with removal volumes contingent upon projected emission reduction pathways (Minx et al., 2018; Schleussner et al., 2022). Therefore, so-called carbon dioxide removal (CDR) technologies such as bioenergy with carbon capture and storage (BECCS) or direct air carbon capture and storage (DACCS) play an increasingly important role in climate science modelling and, more

recently, climate policy, including in the EU (Schenuit et al., 2021). However, CDR technologies such as BECCS or DACCS do not yet exist at scale – and it is uncertain whether they can ever be scaled in a sustainable way to the extent projected in many mitigation pathway models (Beck and Oomen, 2021; Larkin et al., 2018). This has prompted major concerns among climate scientists that expectations of future large-scale availability of CDR technologies may provide an unwarranted justification for deterring mitigation and decarbonization, severely exacerbating climate change if CDR technologies will fail at scale – an effect which has been discussed as a moral hazard or mitigation deterrence (Carton et al., 2023; Grant et al., 2021; Lenzi, 2018; Markusson et al., 2018).

However, while such mitigation deterrence effects have been observed in climate model results, systematic empirical evidence of

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whether and, if so, how expectations of CDR technologies defer or undermine efforts to reduce GHG emissions in climate policy is lacking (Stoddard et al., 2021, p. 668; Beck and Oomen, 2021; Lenzi, 2018). This is a critical knowledge gap, since understanding how eventual instances of CDR-related mitigation deterrence occur – i.e., through what policy designs, actors, institutional mechanisms, imaginaries and forms of knowledge – is crucial to prevent or at least minimize such effects in future climate policy-making. Conversely, deepening our knowledge on how CDR-related mitigation deterrence can be prevented or limited is crucial to make sure that worries of moral hazard or mitigation deterrence do not forestall the development and deployment of CDR technologies at the scale and pace required to avert hazardous climate change (i.e., a potential ‘removal deterrence’).

This contribution therefore seeks to develop a conceptual approach to systematically study mitigation deterrence effects related to the integration CDR technologies into climate policy-making. Our approach integrates literature on sociotechnical imaginaries and the cultural political economy of mitigation deterrence with historical-materialist policy analysis. It follows in-case process tracing to uncover whether and how expectations of CDR technologies either weaken ‘conventional’ mitigation targets in policy formulation or encourage policy designs which treat ‘conventional’ mitigation and CDR as equivalent, thereby facilitating the offsetting of emissions which could (and should) be abated by ‘conventional’ decarbonization efforts. We apply this approach to the integration of CDR technologies into EU climate policy, focusing in particular on the current negotiations over the EU carbon removal certification framework which aims at turning the EU into a frontrunner in CDR climate policy integration.

Section 2 situates our approach in the context of the relevant literature on moral hazard and mitigation deterrence effects related to climate engineering in general and CDR technologies more specifically. Section 3 then introduces our approach to study mitigation deterrence in climate policy-making processes, which we subsequently apply to EU climate policy in Section 4. Section 5 draws some important conclusions from our analysis of these initial steps of CDR climate policy integration and derives policy recommendations from our findings. Our main contention is that even though it is too early to pin-point clear-cut instances of mitigation deterrence related to CDR technologies in EU climate policy-making, critical opportunities to prevent such effects have already been missed in both climate target formulation and policy design.

2. Mitigation deterrence – moving from the abstract to the concrete

While in recent times, the concept of negative emissions has gained prominence in climate science and policy with the proliferation of net zero targets (Hale et al., 2022), debates about carbon removal are not entirely new. Already in the late 1970 s CDR was conceived as a substitute for decarbonization (Carton et al., 2023), and the inclusion of carbon sinks in the Kyoto protocol was motivated by the quest for convenient alternatives to emission reductions, based on a constructed fungibility of carbon removals and emissions reductions (Carton et al., 2020). However, carbon uptake in terrestrial and aquatic ecosystems is limited and reversible, particularly as many ecosystems are vulnerable to heat stress and other weather extremes set to increase with rising global temperature levels (Keenan and Williams, 2018; Mackey et al., 2013). The specific allure of CDR technologies, particularly DACCS, therefore, lies in the promise that they are much more scalable than ecosystem-based CDR, and that CCS-based geochemical carbon storage is near-permanent. This exacerbates the problem of implied fungibility between removals and emission reductions, giving rise to serious concerns in the academic debate that expectations of CDR technologies may instigate moral hazard and delay emission reductions (McLaren et al., 2019).

Informed by behavioral economics, most contributions on moral

hazard related to climate engineering focus on perverse incentives, biased perception and lack of information on behalf of individual decision-makers (even though some authors have taken into account psychological factors and arguments from moral philosophy) (Gardiner, 2011; Morrow, 2014). In so doing, these contributions tend to conceptualize technologies as independent of social relations and material interests. Therefore, as Markusson et al. (2018) have argued, these approaches are significantly limited in their capacity to understand the potential role of CDR technologies to delay or impede efforts to reduce GHG emissions in climate policy-making processes.

The concept of mitigation deterrence, by contrast, is rooted in a cultural political economy framework (Sum and Jessop, 2013) and aims to capture the possible allure of CDR technologies as sociotechnical imaginaries and strategies reflecting dominant logics of techno-fixes and material interests (Low and Boettcher, 2020; Markusson et al., 2022). This approach enables an understanding of the potential role of CDR technologies as part of contested attempts to delay decarbonization – specifically, by functioning to sustain high-level policy conclusions that climate targets can be achieved without disruptively transforming global energy and economic systems (Carton et al., 2023; Markusson et al., 2018, p. 2). As such, mitigation deterrence through CDR is part of a broader spectrum of discourses and strategies of climate delay (Lamb et al., 2020) and climate obstruction (Plehwé et al., 2023) that perpetuate carbon lock-in (Unruh, 2000), but is at the same time specific in that CDR – by acting directly on GHG concentrations – promises to be a substitute for emission reductions and decarbonization tout court.

However, while these contributions have outlined a sophisticated theoretical understanding of mitigation deterrence, the debates around mitigation deterrence have taken place a relatively abstract level so far. Whether, and if so, how, mitigation deterrence related to CDR technologies actually occurs in climate policy-making processes has not yet been investigated. Conversely, Geden, Schenuit and colleagues (Geden et al., 2019; Schenuit et al., 2021; Schenuit and Geden, 2022) have pioneered research on the actors, positions and politics which underpin the integration of CDR technologies into EU climate policy, highlighting looming conflicts between economic sectors and member states over the distribution of residual emissions and responsibilities for carbon removals, but have not yet focused on eventual mitigation deterrence effects related to CDR technologies in EU climate policy.

3. Chasing mitigation deterrence through policy analysis

To empirically uncover mechanism of mitigation deterrence, we suggest an approach to policy analysis based on within-case process tracing and different strands of critical policy studies. Specifically, we combine an emphasis on the social production of knowledge drawn from interpretative approaches (Fischer et al., 2015) – focusing, in particular, on sociotechnical imaginaries (Beck et al., 2021; Jasanoff and Kim, 2015) and public policy paradigms (Carson et al., 2009) – with historical-materialist approaches to policy analysis (Brand et al., 2022; Haas, 2019). Through this, we aim to understand not only how socio-technical imaginaries and policy paradigms constitute and justify policy trajectories, but also what actors and related material interests shape the integration of CDR technologies into EU climate policy on the ground.

The concept of sociotechnical imaginaries permits a capturing of relatively coherent collective “visions of desirable futures”, underpinned by specific ideas and expectations regarding future society and technological progress and how they mutually enable and determine each other (cf. Beckert, 2016; Jasanoff, 2015, p. 4). Following the notion of co-production (Beck et al., 2021), the way a phenomenon such as CDR technology is represented in specific imaginaries can be assumed to shape the way it is acted upon, e.g., it motivates and justifies particular types of CDR governance and, more generally, specific climate policies. Crucially, alternative imaginaries privilege or exclude certain social actors, forms of action and policy choices from consideration, which is why the analysis of the emergence, variation, selection and retention of

competing imaginaries in policy processes is essential. The policy paradigm concept refers to the forms of knowledge, the distribution of competences and responsibilities, and the forms of participation that are deemed possible, appropriate and legitimate on behalf of key policy-makers (Carson et al., 2009; Geden et al., 2018). We assume that dominant (climate) policy paradigms are sustained by particular socio-technical imaginaries, but can also be challenged and eventually modified or replaced through the emergence and consolidation of new sociotechnical imaginaries.

We combine these concepts with historical-materialist approaches to policy analysis (HMPA) which have a distinct focus on the granular examination of a broad range of state and civil-society actors, and how their respective material interests and strategies are mediated through the state as a strategic-selective terrain (Brand et al., 2022; Haas, 2019). We follow the three-step approach of context, process and actor analysis developed in HMPA to reveal the main policy trajectories as well as the constellation of actors and interests regarding the integration of CDR technologies into EU climate policy and to chase eventual mitigation deterrence effects. While we recognize that also ecosystem-based CDR (e.g., forestry) is no less technological than other CDR approaches, our focus is specifically on the role of BECCS and DACCS due to their specific characteristics highlighted above, i.e., alleged availability and permanency, and because these two approaches are explicitly considered by the European Commission.

- (1) *Context analysis* reconstructs the historical and political background against which CDR technologies are integrated into EU climate policy and identifies the overarching development trajectories that have underpinned, shaped and driven these processes. It takes into account the contested role of CDR technologies in the IPCC assessment reports and international climate policy more broadly (Beck and Mahony, 2018) influencing perceptions and expectations of CDR technologies in the EU policy context, as well as the multilevel and polycentric architecture of EU climate policy (Rayner and Jordan, 2013; Wurzel et al., 2019) and the prevailing EU climate policy paradigm with its focus on emission reduction efforts and green growth (Geden et al., 2018).
- (2) *Process analysis* then investigates how specific CDR policy options and related sociotechnical imaginaries find their way into the crucial institutional terrains of policy-making such as the European Commission. The goal is to uncover the selectivities of these terrains and dominant policy paradigms which underpin policy elaboration, i.e., mechanisms of differentiation that privilege some forms of expert knowledge (e.g., specific mitigation models) and underlying imaginaries as well as access of some actors over others (Beck et al., 2021; Jessop, 1990). Against this background, and focusing on decisive moments of contestation and turning points in the policy formulation, negotiation and decision-making process across the institutional terrains of policy-making (within the European Commission, in the European Parliament, the European Council and the Council of Ministers as well as the formal and informal trilogue process), it can be examined which policy options were selected over possible alternatives, which compromises emerged, and how and why this occurred.
- (3) *Actor analysis*, in parallel to the process analysis step, identifies key actors and their strategies as well as related interests. Following HMPA, we adopt a broad perspective on policy-relevant actors that can be expected to take position towards CDR technologies during the ongoing integration and ‘normalization’ of CDR technologies in EU climate policy. This includes not only public institutions (e.g., ministries) and parties, but also business actors belonging associated with different class fractions (e.g., agribusiness, bioeconomy, the fossil capital, emissions intensive non-energy industries and green economy businesses)

as well as civil society actors such as NGOs and think tanks. The goal is to reveal which actors are the main driving force behind the integration of CDR technologies into EU climate policy, why and which approaches to CDR they promote, and which actors oppose or obstruct this integration. This permits an assessment of emerging alliances, common interests and fault lines in the politics of CDR in the EU (between, but also traversing, member states).

This three-step approach provides the basis for chasing mitigation deterrence through within-case process tracing (Beach, 2016). While mitigation deterrence certainly is a highly elusive phenomenon that can occur in multiple, albeit highly subtle ways which are notoriously difficult to pin down, we expect that it can happen in policy-making processes in essentially two traceable ways.³ First, policy-makers, informed by specific sociotechnical imaginaries and seeking to mediate competing material interests and forge compromises, may formulate less ambitious or less clearly defined emission reduction targets. In so doing, they rely on the assumption that due to the future large-scale availability of CDR technologies such reduction is less urgent or does not have to be as comprehensive. According to this assumption, less aggressive emission reduction in the short- and mid-term (i.e. in the 2020 s, 2030 s and 2040 s) and a corresponding overshooting of the CO₂ budget can be compensated for by net-negative emissions later in the 21st century, and/or long-term abatement of emission sources does not have to be as far-reaching because residual emissions can be amply

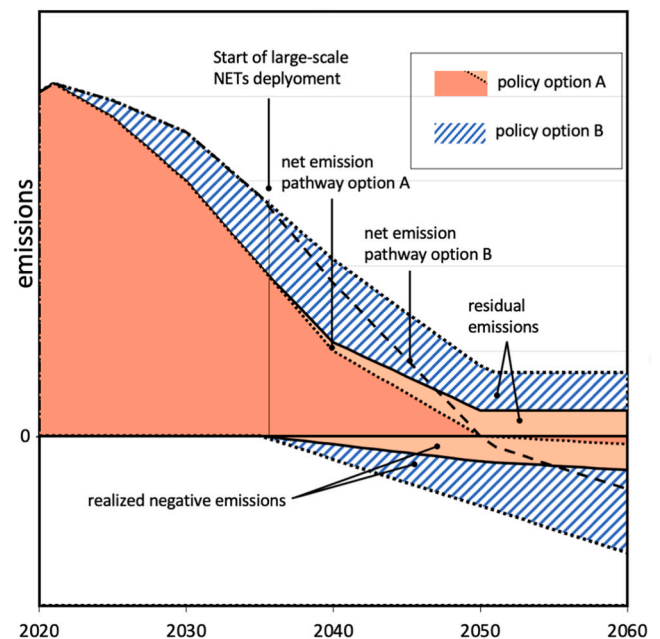


Fig. 1. Illustrative mitigation scenarios for two stylized policy options with low and high mitigation deterrence potential. Note: Policy option A foresees relatively low CDR technology deployment and separates targets for emission reduction and removals (represented as solid lines). Policy option B, by contrast, does not separate these targets and assumes large-scale future CDR technology deployment, allowing for higher residual emissions and less aggressive emission reductions, compensated by higher net negative emissions after 2050.

³ In line with the relevant literature on mitigation deterrence (Markusson et al., 2018), we here understand mitigation deterrence as reduced or delayed mitigation in terms of ‘conventional’ mitigation measures, i.e. (at-source) emission reduction.

balanced with CDR technologies in the future (see Fig. 1 for a stylized representation of two diverging policy options). Secondly, policy-makers may resort to climate policies that treat emission reductions and carbon removals through CDR technologies as equivalent in reaching climate targets (e.g., unrestricted integration of carbon removal credits generated by CDR technologies into carbon trading), relying on the assumption that reductions and removals are equivalent in terms of their climate effects and thus entirely fungible.

Importantly, according to this approach to mitigation deterrence in climate policy-making, the mere integration of CDR technologies into climate policy does not automatically constitute mitigation deterrence. Instead of weakening emission reduction targets, technology-based carbon removal could simply supplement emission reductions, or climate policies may incentivize the scale-up of CDR technologies without treating removals as equivalent to emission reductions, thereby ensuring that such removals do not substitute emission reduction efforts (for a detailed discussion cf. conclusion).

Based on this understanding, we suggest a two-step approach to trace mitigation deterrence in climate policy-making. First, as part of the process analysis step (see above), the range of policy options considered by the relevant policy-makers needs to be systematically assessed regarding (a) the level of ambition and specificity of at-source emission reduction targets (i.e., scope and pace of emission reductions), and/or (b) the degree of assumed equivalence between emission reduction and removals. Second, the goal is to find out whether and how the expectations regarding the future large-scale availability of CDR technologies has affected the policy formulation along dimensions (a) and (b), and which sociotechnical imaginaries, actors and institutional selectivities were critical in this regard. In other words, the task here is to identify, through within-case process tracing, causal mechanisms (or the absence thereof, cf. Beach, 2016) between expectations and sociotechnical imaginaries of CDR technologies and how they are mobilized by different actors on the one hand and the variation, selection and retention of climate policy options on the other hand.

To be sure, this is not to say that mitigation deterrence effects can only occur in the ways described above. For instance, the efficacy of specific sociotechnical imaginaries of CDR technologies and/or dominant policy paradigms may imply that certain policy options are never seriously considered in the policy process, or not even articulated in the first place. However, such forms of mitigation deterrence, even though real, are hardly empirically traceable when studying policy-making processes.⁴ Therefore, this narrow approach – given the controversy whether mitigation deterrence or moral hazard effects exist at all (Jebari et al., 2021) – appears to us ultimately more robust and cogent. We also assume – and indeed observe in the empirical material – that involved policy-makers and stakeholders are at least to some extent aware of the issue of mitigation deterrence in the formulation and negotiation of policies. This implies a broader range of policy options and allows us to investigate whether and, if so, how precautions or safeguards to preempt mitigation deterrence were considered, implemented or rejected in the policy-process (e.g., the separation of removal and emission reduction targets).

In the remaining part of this paper, we focus on a tentative application of this approach to the Sustainable Carbon Cycle communication and the ongoing negotiations over the EU's Carbon Removal Certification Framework (CRCF). Methodologically, we used three semi-structured expert interviews with a policy-consultant advising the

Commission in drafting the certification proposal, a close scientific observer of CDR policy integration and an NGO representative as a entry point for an extensive qualitative document analysis. Besides official EU documents relevant for CDR policy we systematically reviewed relevant reports as well as publicly available statements (e.g., press releases) by actors (e.g., environmental NGOs, business associations, advocacy groups, think tanks), identified through the expert interviews as well as web search. In addition, we collected and screened the stakeholder responses in the public consultation on the CRCF (231 submissions). Based on this material we focused on those documents that were particularly informative or revealing with respect to conflicts over possible mitigation deterrence effects in the policy-making process, specifically searching for instances where stakeholders raised explicit criticism, urged substantial policy changes, or made additional demands.

4. CDR policy integration and the EU carbon removal certification framework – a case of mitigation deterrence?

Potential for mitigation deterrence exists in a variety of ongoing and future policy-making processes in EU climate policy, particularly the negotiation, formulation and structure of the EU's 2040 climate target as well as the integration of CDR technologies into the three main regulatory pillars of EU climate policy: the EU Emission Trading Scheme (EU ETS), the Effort Sharing Regulation (ESR) and the land-use, land-use change and forestry (LULUCF) regulation. However, as the pending integration of CDR technologies into these main pillars is still pending, we here focus on the Sustainable Carbon Cycles strategy and the Carbon Removal Certification Framework (CRCF) – the most advanced and concrete step towards CDR technology policy integration in the EU to date. Nonetheless, to situate the conflicts around the CRCF in the broader context of EU climate policy, we first briefly recapitulate the three main pillars of EU climate policy, the institutional set-up of EU climate policy-making, its dominant policy paradigm and how current policy trajectories may result in future mitigation deterrence effects.

4.1. Setting the scene: EU climate policy pillars and potentials for mitigation deterrence

EU climate policy has mainly evolved around GHG emission reduction targets that have become increasingly ambitious over the last two decades, currently aiming at a 55 per cent reduction by 2030, climate neutrality by 2050 and net-negative emissions thereafter (Oberthür and von Homeyer, 2023). To achieve these targets, EU climate policy relies on three major pillars. First, the EU ETS established in 2003 and operational since 2005 a carbon price for large industrial and power plants as well as (since 2008) domestic aviation. A further extension to maritime transport as well as a second ETS covering buildings, road transport and fuels was negotiated under the Fit for 55 Package as an instrument intended to help Member States meet their ESR commitments (see below). ETS reform under the Fit for 55 Package also aims at a faster reduction of allowances as carbon prices in the ETS were extremely low and thus largely ineffective in the past due to an excess of allowances in the market (Jevnaker and Wettstad, 2017).

The second pillar, the ESR, was established in 2002 and determines the distribution of emission reductions between member states along national targets in those sectors which have not been covered by the ETS (i.e., road transport, buildings, small industrial installations, agriculture and waste) along differentiated emission reduction targets across member states (Oberthür and von Homeyer, 2023, p. 452). Based on the 2030 Climate and Energy Policy Framework, a the third pillar, the LULUCF Regulation (LULUCF Regulation, 2018), was introduced relatively late in 2018, covering both GHG emissions and removals from land-use, land-use change and forestry (building on previous efforts by Member States to account for LULUCF sinks as part of their commitment to the Kyoto Protocol) (Savarese et al., 2020). With the proposal of the Fit for 55 Package in 2021 (European Commission, 2021a), the

⁴ This does not imply that it is impossible to study mitigation deterrence effects outside of policy-making processes, as literature on mitigation deterrence or obstruction in climate modelling has amply demonstrated. For instance, introducing CDR technologies into Integrated Assessment Models increases near-term emissions as compared to scenarios which do not rely on CDR technologies, allowing for less stringent near-term climate action (cf. Carton et al., 2023; Lenzi, 2018).

Commission currently aims at implementing the 55 per cent emission reduction target by 2030 and the climate neutrality by 2050 target which are enshrined in the European Climate Law as part the European Green Deal (European Climate Law, 2021; European Commission, 2019) (Fig. 2).

EU climate policies are shaped and need support from the three main EU institutions: Proposals are formulated by the Commission and adopted by the Council of Ministers (through qualified-majority voting) and the European Parliament (Jevnaker and Wettstad, 2017). Intra- and interinstitutional negotiations are tied together through the so called informal trilogue process, i.e., ongoing coordination processes between the Commission, the Parliament and the Council to identify positions which can find approval in all three institutions (Héritier and Reh, 2012; Syrovatka, 2022). Also the Commission itself is not a monolithic bloc (Hartlapp et al., 2014), with differences and conflicts occurring between various Directorate-Generals, in climate policy-making particularly between Directorate-General for Climate Action (DG Clima) and other Directorate-Generals (Skovgaard, 2014).

Following Geden et al. (2018) the EU's dominant climate policy paradigm has so far relied on the notion that promoting low-carbon technologies will deliver emission reduction targets while simultaneously stimulating green growth. Fundamental to this approach is the assumption of decoupling, i.e., that economic growth can be reconciled with the reduction of GHG emissions and the use of natural resources (Haberl et al., 2020). Along these lines, the Commission has specifically emphasized the idea of a 'circular economy', i.e., reducing waste output and original resource input by closing resource flow loops (Haas et al., 2015), as highlighted particularly in the Commission's Circular Economy Action Plan (European Commission, 2020a).

Although the EU is generally considered a frontrunner or leader in international climate policy (Oberthür and Groen, 2017) and the Commission was an early advocate for the inclusion of CDR in the IPCC's mitigation scenarios, CDR technologies were not seriously considered in the EU's internal climate policy until the 2018, when the IPCC's Special Report introduced the idea of net zero as a new target structure to international climate policy (Geden and Schenuit, 2020). This target structure was subsequently reflected in the Commission's proposal for a new long-term climate strategy, which also – and for the first time – explicitly mentioned the aim of rolling out CDR technologies in the EU (European Commission, 2018). Remarkably, the 2018 long-term strategy laid out two pathways towards zero emissions in 2050 which differ not only in the projected amount of emissions removed by CDR technologies (notably BECCS and DACCS), but also with regard to the residual emissions which need to be balanced by CDR in 2050 (European Commission, 2018, p. 23). Crucially, the pathway option more heavily based on CDR technologies also allows for more residual emissions for some economic sectors, thereby foreshadowing potential conflicts between economic sectors and member states over the total amount and

the distribution of what will be considered residual emissions (cf. Geden and Schenuit, 2020).

However, the Commission's 2018 proposal for a long-term climate strategy as well as the more recent European Climate Law (European Climate Law, 2021) have left open how CDR technologies should be integrated into the three main regulatory pillars of EU climate policy. Like the other two pillars, the *first* pillar, the EU ETS, which covers almost half of the EU's GHG emissions (Jevnaker and Wettstad, 2017), has so far been kept separate from CDR technologies,⁵ even though first attempts have been made by members of the European People's Party Group in the ENVI Committee of the European Parliament to directly integrate CDR technologies into the ETS as part of the ongoing ETS reform (Interview NGO). The ENVI Committee now voted for a sequenced approach, asking the Commission to present by 2025 a report on how "negative emissions can be covered by emission trading" and how such negative emissions "can be accounted for and incentivized in emission trading" (European Parliament, 2022, p. 17–18). Importantly, the Committee also demands "safeguards to ensure that such removals are not offsetting necessary emissions reductions in accordance with Union climate targets" (ibid.), i.e., to prevent mitigation deterrence effects. Such effects would loom large if carbon removal credits generated by CDR technology providers would be directly tradable within the ETS. In this case, there would be a strong incentive to substitute emission reductions by purchasing removal credits if the price for emission allowances exceeds that for removal credits (on alternative options of a restricted or separated integration cf. Rickels et al., 2022).

The *second* pillar, the ESR has similarly remained separate from CDR technologies so far. What is relevant for eventual future mitigation deterrence effects related to expectations of CDR technologies, however, is the recently established flexibility between the ESR and the *third* pillar of EU climate policy, the 2018 LULUCF regulation. Covering both emissions *and* removals from land-use, land-use change and forestry, the LULUCF regulation centers on the so called no-debit-rule, i.e., the commitment that emissions in this sector do not exceed removals in the period from 2021 to 2030⁶ (Regulation 2018/841, Art. 4). However, if Member States fail to respect the no-debit-rule, they can balance an excess of LULUCF emissions with emission allowances from the ESR (cf. Schenuit and Geden, 2022). Conversely, removals from the LULUCF sector can balance emissions from ESR sectors up to 280 Mt CO₂eq (Savarese et al., 2020, p. 216). While this interpillar flexibility does not cover carbon removals generated by CDR technologies, it might serve as a blue-print for how such removal credits may offset emission reductions under the ESR in the future, which would – depending on the level and scope of flexibility and substitutability – effectively deter mitigation efforts in terms of emission abatement.

The updated climate target of reducing GHG emissions by at least 55 per cent by 2030 enshrined in the EU climate law arguably implies a similar tendency towards future mitigation deterrence effects related to CDR technologies. As already mentioned above, pertinent literature on how to prevent mitigation deterrence related to CDR technologies insists, first and foremost, on a policy of separation between emission reductions and removals, particular in regard to climate policy target formulation and structure (Carton et al., 2021; McLaren et al., 2019). However, with the updated target of a *net* reduction of GHG emissions of 55 per cent by 2030 (compared to 1990 levels), the EU has already made a choice against the strict separation of emission reductions and removals, insofar as this target includes both emission reduction *and*

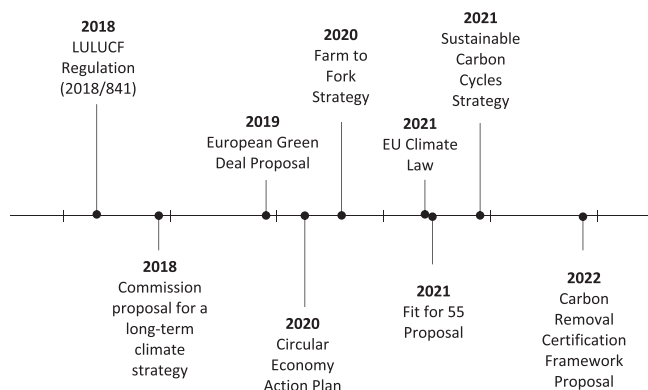


Fig. 2. Key events in EU climate policy since 2018.

⁵ However, the ETS is already relevant for CDR technology development in that EU Innovation Fund, which is financed from the proceeds of the ETS emission allowances, currently funds several pilot projects CDR and carbon capture and utilization technologies (European Commission, 2023).

⁶ As part of the Fit for 55 reform the EU institutions recently agreed that as of 2026 removals should start to exceed emissions, i.e., to go beyond the no-debit-rule from 2026 onwards (Simon, 2022).

removals from the LULUCF sector (Elkerbout and Bryhn, 2021, p. 14). Crucially, however, the European Parliament insisted that the contribution of the LULUCF sink to the 2030 target is limited to its projected removal potential in 2030 (225 Mt CO₂eq) “in order to ensure that sufficient mitigation efforts are deployed up to 2030” (European Climate Law, 2021, Art. 4(1); cf. Keating, 2022), and removals from CDR technologies cannot (yet) be accounted for under the 2030 target. Nonetheless, effectively counteracting mitigation deterrence effects through CDR technologies will require an overhaul of the current climate target structure by separating out emission reductions and removals in the 2040 target, or at least by clearly limiting and delineating the contribution of CDR technologies in the pathways towards a 2040 net target. Even if targets are clearly separated, however, projected removal trajectories may serve as a justification to relax emission reduction targets, particularly as target formulation is an iterative process (CPES 2021, p. 5) – the upcoming negotiation on the 2040 target are therefore a highly critical case to watch.

4.2. The EU carbon removal certification framework – tracing a trajectory of forestalled safeguards

So far, given the tentative stage of CDR integration in EU climate policy, we have mainly highlighted potentials for future mitigation deterrence related to CDR technologies in the three core pillars of EU climate policy. We now turn to the Sustainable Carbon Cycles Strategy and the CRCF, which – even though arguably less relevant than future conflicts over how to integrate CDR technologies into the 2040 target or the EU ETS – provides some important and indeed rather unexpected early insights into policy-related mitigation deterrence risks.

The importance of a carbon removal certification framework is relatively straight-forward: For CDR to be integrated into climate policy, there needs to be a consistent and transparent approach to defining, quantifying and accounting removals, especially given major concerns regarding the permanency or duration of removals and related assumptions about the fungibility of removals and emission reductions (cf. McLaren, 2020). In other words, there needs to be a system of monitoring, reporting and verification to ensure that CDR providers actually remove carbon dioxide from the atmosphere and durably store it, and that also clarifies liability in case removals are reversed (i.e., if CO₂ is released from storage). While a variety of such schemes already exist on voluntary carbon markets, the Commission’s aim is to establish a reliable, common and consistent minimum standards for carbon removals in the EU (Dahm, 2022).

The Commission’s initiative for a certification framework for removals was first revealed in the in the 2020 Circular Economy Action Plan (European Commission, 2020a), subsequently picked up in the so called 2020 Farm to Fork strategy (European Commission, 2020b) and then concretized in the 2021 Sustainable Carbon Cycle communication (European Commission, 2021b; cf. Geden and Schenuit, 2020). This trajectory is relevant insofar as it indicates that the CRCF initiative was – from the very beginning – intertwined with the political project of ‘greening’ the Common Agricultural Policy (Interview Scientist), which the Commission has been pursuing since the early 2010 s (Geden et al., 2018). As a result, the concept of carbon farming (see below) features prominently in the Commissions Sustainable Carbon Cycle communication.

The Sustainable Carbon Cycle communication itself constitutes the first comprehensive attempt by the Commission to kick-off and shape the policy debate about CDR integration, including CDR technologies such as BECCS and DACCS. It marks a significant departure from the hitherto dominant climate policy paradigm in the EU (Geden et al., 2018) insofar as it concedes that ‘conventional’ mitigation in terms of emission reductions alone are no longer sufficient to limit global warming to 1.5 °C (European Commission, 2021b, p. 1–2). The strategy therefore specifically establishes the goals that carbon farming activities increase the EU’s land sink by 42Mt of CO₂eq by 2030, and that another

5Mt of CO₂ “should be removed every year from the atmosphere and permanently stored using technological solutions by 2030” (ibid., p. 9, 17). For the latter aim to be achieved, the strategy also envisages the development an open-access CO₂ transport infrastructure that connects emission sources with (geological) storage sites across the EU and beyond, particularly because “not all Member States have access to suitable storage sites” (ibid., p. 18).

Despite these additional foci on CCS-based removal activities and a CO₂ transport infrastructure, the main emphasis of the strategy lies on carbon farming, defined by the Commission as “a green business model that rewards land managers for taking up improved land management practices, resulting in the increase of carbon sequestration in living biomass, dead organic matter and soils by enhancing carbon capture and/or reducing the release of carbon to the atmosphere” (ibid., p. 4). This peculiar emphasis on carbon farming as the dominant approach to CDR set by the Sustainable Carbon Cycle communication is crucial for the ensuing trajectory of CDR policy integration and related mitigation deterrence risks in at least three respects: *First*, even though the strategy was spearheaded by DG Clima within the Commission, it brought CDR into the orbit of the Directorate-General for Agriculture and Rural Development (DG Agri) and the EU’s agricultural policy, the EU policy area with the largest budget and, correspondingly, highly organized agribusiness interests. This opened the floodgates for intense pressure to turn carbon removal activities into a new revenue stream for the EU’s agricultural sector (Interview Scientist, Interview NGO), as evident particularly in the stakeholder consultation process for the certification of carbon removals. For instance, the German Farmers’ Association (DBV) insisted that generating carbon removals on agricultural soils can only succeed if there is a “long-term additional remuneration”, i.e., financed from additional funds outside of the CAP such as through the EU ETS (DBV, 2022, own translation).

Secondly, the linkage to agricultural policy also implied pressure to broaden the definition of carbon removal. As can be seen from the definition above, carbon farming as understood by the Commission not only includes practices which increase of carbon sequestration and enhance carbon capture, but also – bewilderingly – practices that aim at emission reduction (“reducing the release of carbon to the atmosphere”). As such, the notion of carbon removal in the Commission’s terminology has come to encompass not only removal activities based on long-term geological storage, but also ecosystem-based approaches with higher reversal risks, and even soil carbon sequestration and activities that ‘merely’ reduce emissions.

Third, the aim of generating additional incentives to ‘green’ agriculture through CDR also placed voluntary carbon markets in a central position of the Commission’s approach to CDR development and deployment. Along these lines, the Commission asserts that “carbon farming credits should become an additional ‘product’ that land managers can sell”, with the demand coming from, for instance, “food processing companies that want to reduce the carbon footprint in their own value chains” (European Commission, 2021b, p. 4). This corresponds to the Commission’s strategy in other policy areas such as the Capital Markets Union to govern through the market (Braun et al., 2018), i.e., to induce market actors into desired action through regulatory arrangements to achieve its policy goals (in this case the upscaling of CDR activities), particularly given the Commission’s constrained fiscal and budgetary policy space.

However, this strategy is of course prone to mitigation deterrence effects, which the Commission seems to be well aware of: The Sustainable Carbon Cycle communication not only insists that decarbonization is “[f]irst and foremost” to achieve the EU’s climate objectives, but also cautions that it is “crucial to ensure that credits generated through carbon farming do not undermine other mitigation efforts” (European Commission, 2021b, p. 1, 4). And yet, despite this evident reflection of the mitigation deterrence risks, the approach to integrate and scale up CDR – at least initially – predominantly via carbon farming has also shaped the Commission’s proposal for a regulation establishing a CRCF

(European Commission, 2022). Notwithstanding massive criticisms from environmental NGOs (Carbon Market Watch, 2022), the CRCF still aims at providing the regulatory basis for the supply of carbon credits on voluntary carbon markets, where these credits could be used to offset continued emissions. Also, and even though the framework conceptually differentiates between “permanent carbon storage” (i.e., storage for “several centuries”), “carbon farming” and “carbon storage in products”, it still subsumes all these approaches under “carbon removal”, including the “reduction of carbon release from a biogenic carbon pool” as part of carbon farming activities (European Commission, 2022, p. 10).

To be sure, extent to which the CRCF will facilitate mitigation deterrence will ultimately depend on how the framework evolves in the negotiations in the European Parliament, the Council and the trilogue process, and how the final framework will be used. For instance, if removal credits based on the CRCF can be used to compensate for unmet emission reductions under the ESR, or directly purchased in place of emission allowances in the EU ETS (which is not explicitly foreseen but also not explicitly ruled out in the current draft version), this would constitute a clear case of mitigation deterrence. Nonetheless, by taking this initial approach to CDR development and deployment through carbon farming and certification, the Commission has arguably already forestalled some options to limit mitigation deterrence effects, e.g., banning the use of carbon credits to offset avoidable emissions through voluntary carbon markets or confining the use of such credits to, for instance, a reverse state-led auction system that is kept separate from other climate policy instruments to prevent any fungibility between removals and emission reductions (cf. Fridahl et al., 2020).

What, then, explains this approach despite the obvious awareness of mitigation deterrence risks in relevant parts of the Commission? Besides the influence of vested interests highlighted above, it appears that the EU’s dominant climate policy paradigm and the related sociotechnical imaginary of a circular economy have contributed to this specific early trajectory of CDR policy integration. First, the idea of connecting CDR policy integration to creating new green business models follows the dominant policy paradigm of linking climate action to green growth potentials. Thus, while irritating the EU’s climate policy paradigm by calling into questions the EU’s climate policy success narrative, CDR policy integration is simultaneously shaped by its continued focus on creating green business models and stimulating green growth. Second, the circular economy imaginary that underpins the EU’s green growth conception similarly pervades the EU’s approach to CDR policy integration: By framing CDR technologies as part of carbon cycles as a key element of the Earth’s system – a notion which detaches CDR technologies from controversial imaginaries of geo- or climate engineering – the EU’s approach to integration CDR technologies into climate policy is now compounded by an extremely loose notion of carbon removal which does not clearly separate between permanent geological storage and ecosystem-based CDR.

A further explanation for this specific approach to CDR policy integration lies DG Clima’s peculiar strategy of CDR development and policy integration (Interview Scientist, Interview NGO): First, DG Clima obviously attempts to use voluntary carbon markets as a test ground to develop its methodologies to monitor and account for carbon removals in order to maintain and expand its global regulatory leadership in the area of carbon removal certification. Second, the emphasis on carbon farming also indicates a strategy of sequencing, i.e., integrating CDR into EU climate policy through land-based approaches which are framed as natural or nature-based solution (Osaka et al., 2021) and thus less controversial than CCS-based CDR first, and only in a second step pro-actively scaling up CDR technologies such as BECCS and DACCS once the limited removal potential of land-based sinks will be widely recognized. Whether this strategy will be successful, however, is far from certain, as the current design of the Sustainable Carbon Cycle communication and the CRCF have already triggered a major backlash against the integration of CDR into EU climate policy among environmental NGOs (Biofuelwatch et al., 2022).

Regarding the constellation of actors and interests, the conflicts over the CRCF reveal a somewhat unexpected picture. Even though fossil capital clearly has an obvious interest in promoting CDR technologies and climate engineering more generally as a way to enable continued fossil use of fuel and to slow down the rate of decarbonization (Carton, 2019; Surprise and Sapinski, 2022), the main impetus for the CRCF and particularly its emphasis on carbon farming was heavily asserted by agribusiness and the biomass sector in an attempt to tap into new sources of revenue. This is not to say, however, that fossil capital has not played a role in CDR policy integration in the EU. One of the key civil society actors which has sought to influence the design of the CRCF is the Zero Emissions Platform (Interview Policy Adviser), with a strong presence of the oil and gas industry among its members (e.g., BP, eni, Shell, or Total). Nonetheless, given the obvious complementarity between its accumulation strategies and CDR, fossil capital has so far taken a rather unobtrusive approach to CDR policy integration to ensure that CDR is not associated solely with the vested interests of the oil and gas industry (Interview NGO).

While both fossil capital and agribusiness share an interest in promoting CDR in general, their interests are set to diverge further down the road on the question of which CDR approach is given priority in political and financial support: Whereas agribusiness will seek to defend carbon farming as the primary focus for CDR policy and financial support, fossil capital but also CDR technology providers will have an interest to shift the focus towards CCS-based approaches. Remarkably, a similar dividing line is also emerging among within a third group of actors comprising environmental NGOs which have so far taken a rather skeptical approach towards CCS-based CDR policy integration: While some NGOs such as BUND (Friends of the Earth Germany) continue to fundamentally oppose CCS and CCS-based CDR technologies (BUND, 2022), others such as Carbon Market Watch have shifted towards supporting CDR approaches combined with long-term storage potentials such as CCS instead of ecosystem-based removal approaches with high risk of reversal, e.g., soil carbon sequestration (Carbon Market Watch, 2022).

5. Conclusion

In this paper, we introduced a novel approach to study mitigation deterrence through CDR technologies in climate policy-making processes. As demonstrated in the tentative application to the EU’s CRCF, this approach allows us to identify the actors and their interests as well as the sociotechnical imaginaries and policy paradigms which shape specific climate policy trajectories. While we could not identify clear-cut instances of mitigation deterrence at this early stage of CDR technology climate policy integration, the analysis reveals that critical opportunities to prevent mitigation deterrence effects have already been missed. This relates in particular to the Commission’s decision against explicitly limiting the use of removal certificates to prevent offsetting of avoidable emissions on voluntary carbon markets, its inadequate definition of what counts as carbon removal, as well as the failure to separate emission reductions and removals in the 55 per cent reduction target in the European Climate Law. In terms of policy recommendations, these findings suggest that the integration of CDR into climate policy should not only rest on a clear separation of removals and emission reductions in climate action target design (McLaren et al., 2019), but also on confining the use of removal credits exclusively to achieving this separate removal target (instead of offsetting emission reductions). Moreover, we also highlight the pitfall of detaching CDR technologies from controversial geo-engineering by framing them as part of seemingly natural carbon cycles, as this runs the risk of blurring the distinction between different forms of CDR with short- and long-term storage, with only near-permanent CDR forms being an adequate instrument to achieve removal targets which complement (rather than substitute) emission reduction targets.

Further research on mitigation deterrence in EU climate policy will

have to focus on the negotiation, formulation and structure of the EU's 2040 climate target as well as the integration of CDR technologies into the three main regulatory pillars of EU climate policy: the EU ETS, the Effort Sharing Regulation and LULUCF Regulation. This will require specific attention to how the key sociotechnical imaginaries identified here have been linked to CDR technologies, how they shape decisions of key policy-makers on the ground, and how the nascent constellation of actors and interests regarding the role of CDR technologies in EU climate policy will consolidate and evolve over time.

To scale up CDR to the required levels in a responsible manner EU climate policy should, first, clearly separate targets for emission reduction from carbon removal objectives with respective separate timetables. Secondly, removal certificates should not be eligible for offsetting emissions in carbon trading to avoid further delays in decarbonization, particularly in high emitting and hard-to-abate sectors (McLaren et al., 2019). Instead, CDR could be incentivized through a reverse state-led auction system (Fridahl et al., 2020), financed, for instance, through revenues from the ETS. This would provide the necessary demand-pull, which together with targeted industrial policy interventions could enable the transition from piloting to scaled operations. Thirdly, such a scheme would have to rely on strict measuring, reporting, verification and accounting of removals (Honegger et al., 2021) focus exclusively on near-permanent CDR options to avoid any confusion with CCU that merely delays emissions (Schenuit et al., 2023), and adequately considers negative environmental and social impacts of different CDR technologies, both within the EU and globally. Finally, a responsible and forward-looking approach to integrate CDR technologies into EU climate policy requires an open debate regarding the required level of CDR by 2050 and beyond. This pertains, on the one hand, to the so far largely “unexplored politics of residual emissions” (Lund et al., 2023), i. e., the question which economic activities we legitimately consider as socially necessary but technically impossible to abate – a question which ultimately also hinges upon on how we assess absolute emission reductions in production and consumption along various demand-side climate change mitigation options (Creutzig et al., 2018). This also relates, on the other hand, to what extent the EU is ready and willing to take responsibility to compensate for its historical “atmospheric appropriation” (Fanning and Hickel, 2023; Mohan et al., 2021) by deploying CDR technologies at a large scale.

CRedit authorship contribution statement

Alina Brad: Conceptualization, Writing – original draft, Writing – review & editing. **Etienne Schneider:** Conceptualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Alina Brad reports financial support was provided by Austrian Science Fund.

Data Availability

Data will be made available on request.

References

- Beach, D., 2016. It's all about mechanisms – what process-tracing case studies should be tracing. *N. Political Econ.* 21, 463–472. <https://doi.org/10.1080/13563467.2015.1134466>.
- Beck, S., Mahony, M., 2018. The politics of anticipation: the IPCC and the negative emissions technologies experience. *Glob. Sustain.* 1, e8 <https://doi.org/10.1017/sus.2018.7>.
- Beck, S., Oomen, J., 2021. Imagining the corridor of climate mitigation – what is at stake in IPCC's politics of anticipation? *Environ. Sci. Policy* 123, 169–178. <https://doi.org/10.1016/j.envsci.2021.05.011>.
- Beck, S., Jasanoff, S., Stirling, A., Polzin, C., 2021. The governance of sociotechnical transformations to sustainability. *Curr. Opin. Environ. Sustain.* 49, 143–152. <https://doi.org/10.1016/j.cosust.2021.04.010>.
- Beckert, J., 2016. Imagined futures: fictional expectations and capitalist dynamics. Harvard University Press, Cambridge, Massachusetts.
- Biofuelwatch, BUND / Friends of the Earth Germany, CCFD - Terre Solidaire, CEO - Corporate Europe Observatory, CIEL - Center for International Environmental Law, Corporate Accountability, European Coordination Via Campesina, Fern, Food & Water Action Europe, Friends of the Earth Europe, Friends of the Earth International, Global Forest Coalition, Heinrich Böll Stiftung, IATP - Institute for Agriculture & Trade Policy, ReCommon, 2022. To avoid the worst effects of climate chaos, we must radically transform systems and achieve Real Zero [WWW Document]. URL (<https://www.realsolutions-not-netzero.org/real-zero-europe>) (Accessed 2.23.23).
- Brand, U., Krams, M., Lenikus, V., Schneider, E., 2022. Contours of historical-materialist policy analysis. *Crit. Policy Stud.* 16, 279–296. <https://doi.org/10.1080/19460171.2021.1947864>.
- Braun, B., Gabor, D., Hübner, M., 2018. Governing through financial markets: towards a critical political economy of Capital Markets Union. *Compét. Change* 22, 101–116. <https://doi.org/10.1177/1024529418759476>.
- BUND, 2022. BUND gegen Kohlendioxid-Deponien im Meer oder an Land. (<https://www.bund.net/service/presse/pressemitteilungen/detail/news/bund-gegen-kohlendi-oxid-deponien-im-meer-oder-an-land-ampel-darf-auch-klimaschaedlichen-plaenen-der-industrie-fuer-landesweite-co2-pipelines-und-exportinfrastruktur-nicht-nachgeben/>) (Accessed 23.02.2023).
- Carbon Market Watch, 2022. What's the use? European Commission messes up definition and utility of carbon removals - Carbon Market Watch. (<https://carbonmarketwatch.org/2022/12/09/whats-the-use-european-commission-messes-up-definition-and-utility-of-carbon-removals/>) (Accessed 23.02.2023).
- Carbon Market Watch, Environmental Coalition on Standards, European Environmental Bureau, Bellona Europa, Institute for Agriculture and Trade Policy, FERN, WWF European Policy Office, 2022. Open Letter RE: Carbon Removal Certification Framework. (<https://carbonmarketwatch.org/wp-content/uploads/2022/11/Open-letter-NGOs-to-Cabinet-on-CRCF-priorities.pdf>) (accessed: 23.02.2023).
- Carson, M., Burns, T.R., Calvo, D., 2009. Paradigms in Public Policy: Theory and Practice of Paradigm Shifts in the EU. Peter Lang, Berlin.
- Carton, W., 2019. “Fixing” climate change by mortgaging the future: negative emissions, spatiotemporal fixes, and the political economy of delay. *Antipode* 51, 750–769. <https://doi.org/10.1111/anti.12532>.
- Carton, W., Asiyambi, A., Beck, S., Buck, H.J., Lund, J.F., 2020. Negative emissions and the long history of carbon removal. *WIREs Clim. Change* 11. <https://doi.org/10.1002/wcc.671>.
- Carton, W., Lund, J.F., Dooley, K., 2021. Undoing equivalence: rethinking carbon accounting for just carbon removal. *Front. Clim.* 0 <https://doi.org/10.3389/fclim.2021.664130>.
- Carton, W., Hougaard, I., Markusson, N., Lund, J.F., 2023. Is carbon removal delaying emission reductions? *WIREs Clim. Change*. <https://doi.org/10.1002/wcc.826>.
- Creutzig, F., Roy, J., Lamb, W.F., Azevedo, I.M.L., Bruin de Bruin, W., Dalkmann, H., Edelenbosch, O.Y., Geels, F.W., Grubler, A., Hepburn, C., Hertwich, E.G., Khosla, R., Mattauch, L., Minx, J.C., Ramakrishnan, A., Rao, N.D., Steinberger, J.K., Tavoni, M., Ürges-Vorsatz, D., Weber, E.U., 2018. Towards demand-side solutions for mitigating climate change. *Nat. Clim. Change* 8, 260–263. <https://doi.org/10.1038/s41558-018-0121-1>.
- Dahm, J., 2022. Commission carbon farming plans leave key questions open [WWW Document]. www.euractiv.com/section/agriculture-food/news/commission-carbon-farming-plans-leave-key-questions-open/ (Accessed 2.23.23).
- DBV, 2022. Kurzstellungnahme des Deutschen Bauernverbandes zur Initiative „Certification of carbon removals – EU rules“ [WWW Document]. URL (https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/13172-Zertifizierung-von-Ma%C3%9Fnahmen-zur-Entfernung-von-CO2-EU-Vorschriften/F3254660_de).
- Elkerbout, M., Bryhn, J., 2021. Setting the context for an EU policy framework for negative emissions. CEPS Policy Insights.
- European Climate Law, 2021. Regulation of the European Parliament and of the Council establishing the framework for achieving climate neutrality and amending Regulations (EC) No 401/2009 and (EU) 2018/1999 (‘European Climate Law’).
- European Commission, 2018. A Clean Planet for all. A European strategic long-term vision for a prosperous, modern, competitive and climate neutral economy. COM (2018) 773 final.
- European Commission, 2019. The European Green Deal. COM(2019) 640 final.
- European Commission, 2020a. A new Circular Economy Action Plan. COM(2020) 98 final.
- European Commission, 2020b. A Farm to Fork Strategy for a fair, healthy and environmentally-friendly food system. COM(2020) 381 final.
- European Commission, 2021a. Fit for 55: delivering the EU's 2030 Climate Target on the way to climate neutrality. COM(2021) 550 final.
- European Commission, 2021b. Sustainable Carbon Cycles. COM(2021) 800 final. https://doi.org/10.1163/2210-7975_HRD-4679-0058.
- European Commission, 2022. Proposal for a Regulation of the European Parliament and of the Council establishing a Union certification framework for carbon removals. COM(2022) 672 final.
- European Commission, 2023. Innovation Fund projects [WWW Document]. URL (https://climate.ec.europa.eu/eu-action/funding-climate-action/innovation-fund/innovation-fund-projects_en) (Accessed 2.23.23).
- European Parliament, 2022. Report on the proposal for a directive of the European Parliament and of the Council amending Directive 2003/87/EC establishing a system

- for greenhouse gas emission allowance trading within the Union, Decision (EU) 2015/1814 concerning the establishment and operation of a market stability reserve for the Union greenhouse gas emission trading scheme and Regulation (EU) 2015/757 (COM(2012)0551 – C9-0318/2012 – 2012/0211(COD)).
- Fanning, A.L., Hickel, J., 2023. Compensation for atmospheric appropriation. *Nat. Sustain.* <https://doi.org/10.1038/s41893-023-01130-8>.
- Fischer, F., Torgerson, D., Durnová, A., Orsini, M., 2015. *Handbook of Critical Policy Studies*. Edward Elgar, Cheltenham, UK, Northampton, MA, USA.
- Fridahl, M., Bellamy, R., Hansson, A., Haikola, S., 2020. Mapping multi-level policy incentives for bioenergy with carbon capture and storage in Sweden. *Front. Clim.* 2 (604787) <https://doi.org/10.3389/fclim.2020.604787>.
- Gardiner, S.M., 2011. Some early ethics of geoengineering the climate: a commentary on the values of the royal society report. *Environ. Values* 163–188.
- Geden, O., Schenuit, F., 2020. Unconventional mitigation: carbon dioxide removal as a new approach in EU climate policy. *German Institute for International and Security Affairs*.
- Geden, O., Scott, V., Palmer, J., 2018. Integrating carbon dioxide removal into EU climate policy: Prospects for a paradigm shift. *WIREs Clim. Change* 9. <https://doi.org/10.1002/wcc.521>.
- Geden, O., Peters, G.P., Scott, V., 2019. Targeting carbon dioxide removal in the European Union. *Clim. Policy* 19, 487–494. <https://doi.org/10.1080/14693062.2018.1536600>.
- Grant, N., Hawkes, A., Mittal, S., Gambhir, A., 2021. Confronting mitigation deterrence in low-carbon scenarios. *Environ. Res. Lett.* 16 <https://doi.org/10.1088/1748-9326/ac0749>.
- Haas, T., 2019. Struggles in European Union energy politics: a gramscian perspective on power in energy transitions. *Energy Res. Soc. Sci.* 48, 66–74. <https://doi.org/10.1016/j.erss.2018.09.011>.
- Haas, W., Krausmann, F., Wiedenhofer, D., Heinz, M., 2015. How circular is the global economy?: An assessment of material flows, waste production, and recycling in the European Union and the World in 2005. *J. Ind. Ecol.* 19, 765–777. <https://doi.org/10.1111/jiec.12244>.
- Haberl, H., Wiedenhofer, D., Virág, D., Kalt, G., Plank, B., Brockway, P., Fishman, T., Hausknost, D., Krausmann, F., Leon-Gruhlalski, B., Mayer, A., Pichler, M., Schaffartzik, A., Sousa, T., Strecek, J., Creutzig, F., 2020. A systematic review of the evidence on decoupling of GDP, resource use and GHG emissions, part II: synthesizing the insights. *Environ. Res. Lett.* 15.
- Hale, T., Smith, S.M., Black, R., Cullen, K., Fay, B., Lang, J., Mahmood, S., 2022. Assessing the rapidly-emerging landscape of net zero targets. *Clim. Policy* 22, 18–29. <https://doi.org/10.1080/14693062.2021.2013155>.
- Hartlapp, M., Metz, J., Rauh, C., 2014. *Which Policy for Europe?: Power and Conflict inside the European Commission*. Oxford University Press, Oxford.
- Héritier, A., Reh, C., 2012. Codecision and its discontents: intra-organisational politics and institutional reform in the European Parliament. *West Eur. Polit.* 35, 1134–1157. <https://doi.org/10.1080/01402382.2012.706414>.
- Honegger, M., Michaelowa, A., Roy, J., 2021. Potential implications of carbon dioxide removal for the sustainable development goals. *Clim. Policy* 21, 678–698. <https://doi.org/10.1080/14693062.2020.1843388>.
- Jasanoff, S., 2015. Future imperfect: science, technology and the imaginations of modernity. In: *Dreamscapes of Modernity: Sociotechnical Imaginaries and the Fabrication of Power*. University of Chicago Press, pp. 1–33. <https://doi.org/10.7208/chicago/9780226276663.001.0001>.
- Jasanoff, S., Kim, S.-H., 2015. *Dreamscapes of Modernity: Sociotechnical Imaginaries and the Fabrication of Power*. University of Chicago Press. <https://doi.org/10.7208/chicago/9780226276663.001.0001>.
- Jebari, J., Taiwo, O.O., Andrews, T.M., Aquila, V., Beckage, B., Belaia, M., Clifford, M., Fuhrman, J., Keller, D.P., Mach, K.J., Morrow, D.R., Raimi, K.T., Visioni, D., Nicholson, S., Trisos, C.H., 2021. From moral hazard to risk-response feedback. *Clim. Risk Manag.* 33 <https://doi.org/10.1016/j.crm.2021.100324>.
- Jessop, B., 1990. *State Theory: Putting the Capitalist State in Its Place*. Polity Press, Cambridge.
- Jevnaker, T., Wettestad, J., 2017. Ratcheting up carbon trade: the politics of reforming EU emissions trading. *Glob. Environ. Polit.* 17, 105–124. https://doi.org/10.1162/GLEP_a.00403.
- Keating, D., 2022. Opinion: EU's new 57% target is more about communication than increased ambition. *Energy Monitor*. URL (<https://www.energymonitor.ai/policy/green-deals/opinion-eus-new-57-percent-target-is-more-about-communication-than-increased-ambition/>) (accessed 2.23.23).
- Keenan, T.F., Williams, C.A., 2018. The terrestrial carbon sink. *Annu. Rev. Environ. Resour.* 43, 219–243. <https://doi.org/10.1146/annurev-environ-102017-030204>.
- Lamb, W.F., Mattioli, G., Levi, S., Roberts, J.T., Capstick, S., Creutzig, F., Minx, J.C., Müller-Hansen, F., Culhane, T., Steinberger, J.K., 2020. Discourses of climate delay. *Glob. Sustain.* 3, e17 <https://doi.org/10.1017/sus.2020.13>.
- Larkin, A., Kuriakose, J., Sharmina, M., Anderson, K., 2018. What if negative emission technologies fail at scale? Implications of the Paris Agreement for big emitting nations. *Clim. Policy* 18, 690–714.
- Lenzi, D., 2018. The ethics of negative emissions. *Glob. Sustain.* 1, e7 <https://doi.org/10.1017/sus.2018.5>.
- Low, S., Boettcher, M., 2020. Delaying decarbonization: Climate governmentalities and sociotechnical strategies from Copenhagen to Paris. *Earth Syst. Gov.* 5 <https://doi.org/10.1016/j.esg.2020.100073>.
- LULUCF Regulation, 2018. Regulation (EU) 2018/ of the European Parliament and of the Council of 30 May 2018 on the inclusion of greenhouse gas emissions and removals from land use, land use change and forestry in 2030 climate and energy framework, and amending Regulation (EU) No 525/2013 and Decision No 529/2013/EU.
- Lund, J.F., Markusson, N., Carton, W., Buck, H.J., 2023. Net zero and the unexplored politics of residual emissions. *Energy Res. Soc. Sci.* 98, 103035 <https://doi.org/10.1016/j.erss.2023.103035>.
- Mackey, B., Prentice, I.C., Steffen, W., House, J.I., Lindenmayer, D., Keith, H., Berry, S., 2013. Untangling the confusion around land carbon science and climate change mitigation policy. *Nat. Clim. Change* 3, 552–557. <https://doi.org/10.1038/nclimate1804>.
- Markusson, N., McLaren, D., Tyfield, D., 2018. Towards a cultural political economy of mitigation deterrence by negative emissions technologies (NETs). *Glob. Sustain.* 1 <https://doi.org/10.1017/sus.2018.10>.
- Markusson, N., McLaren, D., Szerszynski, B., Tyfield, D., Willis, R., 2022. Life in the hole: practices and emotions in the cultural political economy of mitigation deterrence. *Eur. J. Futures Res.* 10. <https://doi.org/10.1186/s40309-021-00186-z>.
- McLaren, D., 2020. Quantifying the potential scale of mitigation deterrence from greenhouse gas removal techniques. *Clim. Change* 2411–2428.
- McLaren, D.P., Tyfield, D.P., Willis, R., Szerszynski, B., Markusson, N.O., 2019. Beyond “net-zero”: a case for separate targets for emissions reduction and negative emissions. *Front. Clim.* 1 <https://doi.org/10.3389/fclim.2019.00004>.
- Minx, J.C., Lamb, W.F., Callaghan, M.W., Fuss, S., Hilaire, J., Creutzig, F., Amann, T., Beringer, T., de Oliveira Garcia, W., Hartmann, J., Khanna, T., Lenzi, D., Luderer, G., Nemet, G.F., Rogelj, J., Smith, P., Vicente Vicente, J.L., Wilcox, J., del Mar Zamora Dominguez, M., 2018. Negative emissions—Part I: research landscape and synthesis. *Environ. Res. Lett.* 13 <https://doi.org/10.1088/1748-9326/aabf9b>.
- Mohan, A., Geden, O., Fridahl, M., Buck, H.J., Peters, G.P., 2021. UNFCCC must confront the political economy of net-negative emissions. *One Earth* 4, 1348–1351. <https://doi.org/10.1016/j.oneear.2021.10.001>.
- Morrow, D.R., 2014. Ethical aspects of the mitigation obstruction argument against climate engineering research. *Philos. Trans. R. Soc. A.* 372 <https://doi.org/10.1098/rsta.2014.0062>.
- Oberthür, S., Groen, L., 2017. The European Union and the Paris Agreement: leader, mediator, or bystander? *Eur. Union Paris Agreeem. WIREs Clim. Change* 8. <https://doi.org/10.1002/wcc.445>.
- Oberthür, S., von Homeyer, I., 2023. From emissions trading to the European Green Deal: the evolution of the climate policy mix and climate policy integration in the EU. *J. Eur. Public Policy* 30, 445–468. <https://doi.org/10.1080/13501763.2022.2120528>.
- Osaka, S., Bellamy, R., Castree, N., 2021. Framing “nature-based” solutions to climate change. *WIREs Clim. Change* 12. <https://doi.org/10.1002/wcc.729>.
- Plehwe, D., Haas, T., Neujeffski, M., 2023. Climate Obstruction in the European Union: Business Coalitions and the Technocracy of Delay. In: Brulle, R. (Ed.), *Climate Obstruction in Europe*. Oxford University Press, Oxford, p. forthcoming.
- Rayner, T., Jordan, A., 2013. The European Union: the polycentric climate policy leader? *WIREs Clim. Change* 4, 75–90. <https://doi.org/10.1002/wcc.205>.
- Rickels, W., Rothenstein, R., Schenuit, F., Fridahl, M., 2022. Procure, bank, release: carbon removal certificate reserves to manage carbon prices on the path to net-zero. *Energy Res. Soc. Sci.* 94 <https://doi.org/10.1016/j.erss.2022.102858>.
- Savarese, A., Perugini, L., Chiriaco, M.V., 2020. Making sense of the LULUCF regulation: much ado about nothing? *RECIEL* 29, 212–220. <https://doi.org/10.1111/reel.12332>.
- Schenuit, F., Geden, O., 2022. Carbon Dioxide Removal: Climbing up the EU Climate Policy Agenda 18.
- Schenuit, F., Colvin, R., Fridahl, M., McMullin, B., Reisinger, A., Sanchez, D.L., Smith, S.M., Torvanger, A., Wreford, A., Geden, O., 2021. Carbon dioxide removal policy in the making: assessing developments in 9 OECD cases. *Front. Clim.* 3 <https://doi.org/10.3389/fclim.2021.638805>.
- Schenuit, F., Böttcher, M., Geden, O., 2023. “Carbon Management”: Chancen und Risiken für ambitionierte Klimapolitik.
- Schleussner, C.-F., Ganti, G., Rogelj, J., Gidden, M.J., 2022. An emission pathway classification reflecting the Paris Agreement climate objectives. *Commun. Earth Environ.* 3 <https://doi.org/10.1038/s43247-022-00467-w>.
- Simon, F.A.
- Skovgaard, J., 2014. EU climate policy after the crisis. *Environ. Polit.* 23, 1–17. <https://doi.org/10.1080/09644016.2013.818304>.
- Stoddard, I., Anderson, K., Capstick, S., Carton, W., Depledge, J., Facer, K., Gough, C., Hache, F., Hoolehan, C., Hultman, M., Hällström, N., Kartha, S., Klinsky, S., Kuchler, M., Löfbrand, E., Nasiritousi, N., Newell, P., Peters, G.P., Sokona, Y., Stirling, A., Stilwell, M., Spash, C.L., Williams, M., 2021. Three decades of climate mitigation: why haven't we bent the global emissions curve? *Annu. Rev. Environ. Resour.* 46, 653–689. <https://doi.org/10.1146/annurev-environ-012220-011104>.
- Sum, N.-L., Jessop, B., 2013. *Towards a Cultural Political Economy*. Edward Elgar, Cheltenham, UK; Northampton, MA, USA. <https://doi.org/10.4337/9780857930712>.
- Surprise, K., Sapinski, J., 2022. Whose climate intervention? Solar geoengineering, fractions of capital, and hegemonic strategy. *Cap. Cl.* <https://doi.org/10.1177/03098168221114386>.
- Syrovatka, F., 2022. *Neue Europäische Arbeitspolitik Umkämpfte Integration in der Eurokrise*. Campus, Frankfurt, Main/New York.
- Unruh, G.C., 2000. Understanding carbon lock-in. *Energy Policy* 28, 817–830. [https://doi.org/10.1016/S0301-4215\(00\)00070-7](https://doi.org/10.1016/S0301-4215(00)00070-7).
- Wurzel, R.K.W., Lieferer, D., Di Lullo, M., 2019. The European Council, the Council and the Member States: changing environmental leadership dynamics in the European Union. *Environ. Polit.* 28, 248–270. <https://doi.org/10.1080/09644016.2019.1549783>.